

DSA0602 Data Handling and Visualization

Class Activity – Unit V

Visualization, Uncertainty & Communication

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Q1. Framing Probability for a Rare Surgical Complication

Version A states: “There is a 2% chance of complication.” Version B states: “Out of 100 patients like you, 2 will experience a complication.”

Evaluation

For a layperson, Version B is generally more likely to be understood correctly because it converts an abstract percentage into a natural frequency. Many people find it easier to imagine a group of 100 patients than to interpret what a 2% probability means.

A known pitfall in probability communication is difficulty understanding percentages and denominators. People may hear “2%” but fail to form an accurate mental picture of the risk. For example, a patient might incorrectly interpret 2% as either “almost no risk” or, in an anxious situation, as a much larger risk than it actually represents. The frequency statement makes the denominator visible: approximately 2 people out of 100 similar patients would experience the complication.

Example

Suppose a hospital has evidence from 1,000 comparable operations and 20 patients experienced the complication. The estimated rate is $20/1,000 = 2\%$. Communicating this as “about 2 out of every 100 similar patients” gives the patient an intuitive reference. Another useful comparison is that approximately 98 out of 100 patients would not experience the complication, although the hospital should avoid implying that the outcome is predictable for any particular individual.

It is also important to explain that this is an estimated risk, not a guarantee. The actual outcome for one patient cannot be determined from the population percentage alone.

When to switch framing

I would recommend Version A when the audience is medically trained, when comparing several risks numerically, or when the exact probability is needed for a statistical calculation. For example, a research report might state that the estimated probability is 0.02 because this value can be directly used in a risk model.

For patient communication, the strongest approach is often to combine both forms: “The estimated risk is 2%, which means about 2 out of every 100 patients like you experience this complication.” This gives both statistical precision and intuitive understanding.

Recommendation

The hospital should normally use the natural-frequency explanation first for patient-facing communication and provide the percentage alongside it. This reduces the chance of misunderstanding while preserving numerical precision.

Q2. Static Confidence Band vs. Animated HOP for Rainfall

A weather agency must choose between a static uncertainty band and an animated Hypothetical Outcome Plot (HOP) for tomorrow's rainfall forecast.

Option A – Static confidence/uncertainty band

A static band is fast to read. A viewer can immediately see the central forecast and the surrounding plausible range without waiting for an animation. It is also technically simple, consumes relatively little battery and bandwidth, and works well on a small mobile screen.

For example, if tomorrow's forecast is 20 mm of rain and the uncertainty band extends from 10 to 35 mm, the user can quickly understand that 20 mm is the central estimate while substantially different rainfall amounts remain plausible.

The main risk is interpretation. A general viewer may assume that the band is a guaranteed minimum and maximum, or that every value inside the band has exactly the same probability. A confidence or predictive interval has a statistical meaning that should be explained in plain language.

Option B – Animated HOP

A HOP displays many simulated possible outcomes one after another. Instead of seeing only two boundaries, the viewer sees a collection of plausible futures. This can make uncertainty more intuitive.

For example, an animation might show simulated rainfall amounts such as 12 mm, 18 mm, 24 mm, 15 mm, 31 mm and 20 mm. The viewer sees that the forecast is not a single fixed number and that some outcomes occur more frequently than others.

However, animation can be slower and more demanding on a mobile device. A user may open the app briefly and not watch the full animation. Rapid movement may also distract from the message. Some users could incorrectly interpret the simulated frames as successive actual weather predictions rather than hypothetical outcomes.

Comparison

- Comprehension speed: static band is better.
- Communication of many possible outcomes: HOP is better.
- Risk of misinterpretation: static band is generally lower if clearly labelled.
- Mobile performance and battery use: static band is better.
- Technical implementation: static band is simpler.

Final recommendation

For a general public mobile application, I would select the static uncertainty band as the primary display. A short explanation such as "The shaded region shows plausible rainfall amounts" can prevent common misunderstandings. An optional HOP could be provided for users who tap "Explore possible outcomes."

Recommendation

The weather agency should prioritize the static band for the main forecast screen because speed, clarity, accessibility and technical reliability matter more than advanced uncertainty exploration for the majority of mobile users.

Q3. Startup Revenue Chart: Proportional Ink and Axis Choice

The startup claims that revenue increased from ₹2 lakhs to ₹2 crores over three years. Since ₹2 crores equals ₹200 lakhs, the final revenue is 100 times the initial revenue.

Problem with the non-zero baseline

The chart uses bars with a non-zero baseline and exaggerates the last bar. This violates the principle of proportional ink. In a bar chart, the visible height of a bar should represent the magnitude of the data from a meaningful zero reference.

For example, if the vertical axis begins at ₹150 lakhs, a ₹200-lakh bar has only 50 lakhs of visible height above the baseline. The viewer therefore sees a bar whose visible length represents 50 rather than 200. An initial value of ₹2 lakhs would be almost invisible. The resulting visual difference is driven partly by the chosen baseline rather than only by the data.

Why the linear scale is appropriate

A linear scale is reasonable for revenue because the units have an additive meaning: ₹100 lakhs is twice ₹50 lakhs. The major problem is not the linear scale itself; it is the truncated baseline.

A corrected bar chart should begin at ₹0. This makes the true 100-fold growth visually honest. The first bar will be very small compared with the final bar, but that is an accurate consequence of the data.

Example of a better communication strategy

If the startup wants to communicate the growth trajectory more clearly, it could use two charts: a zero-baseline bar chart for honest magnitude comparison and a line chart with a logarithmic y-axis to show proportional growth over time. The log chart should be explicitly labelled because equal vertical distances represent equal ratios rather than equal rupee increases.

For instance, the jump from ₹2 lakhs to ₹20 lakhs is a 10-fold increase, while ₹20 lakhs to ₹2 crores is another 10-fold increase. A log scale can display these equal multiplicative changes at comparable vertical distances.

Recommendation

The startup should use a zero-baseline bar chart as the primary evidence of revenue magnitude. A clearly labelled logarithmic chart can be added as a secondary analytical view if the purpose is to explain multiplicative growth.

Q4. Rainbow and Red-Green Scales on a COVID Dashboard

The city dashboard uses a rainbow/Jet scale for infection rates and red-green colours for safe versus high-risk areas. Both choices create problems, but the red-green scheme is the more serious public-health concern.

1. Perceptual accuracy

Jet/rainbow is not perceptually uniform. Equal numerical changes can appear to have very different visual importance because the palette moves through strongly contrasting colours. A smooth increase in infection rate can therefore look like it contains artificial thresholds.

For example, infection rates of 40, 45, 50 and 55 cases per 100,000 may be numerically close, but a rainbow palette can make one transition appear much more dramatic simply because the colour changes rapidly at that point. This can cause viewers to infer a boundary that is not supported by the data.

2. Accessibility

Red-green is especially problematic for people with red-green colour-vision deficiency. If a dashboard uses only red and green to distinguish high-risk from safe districts, some users may have difficulty determining which areas belong to which category.

The problem can be reduced by adding text labels, symbols, patterns, or a light-to-dark sequential scale. Colour should not be the only encoding for an important health warning.

3. Public trust

During a health emergency, the visualization must look neutral, consistent and evidence-based.

Arbitrary rainbow transitions can make the map appear dramatic rather than analytical. Red and green also carry strong emotional meanings—danger and safety—which may cause people to interpret small numerical differences as categorical judgments.

A transparent legend with actual infection-rate values helps users understand what the colours mean and can improve trust.

Severity ranking

1. Red-green: more severe harm. It combines an accessibility problem with a potentially consequential safe/high-risk interpretation.
2. Jet/rainbow: serious harm. It can create false visual boundaries and exaggerate some numerical transitions, but it does not create the same specific red-green accessibility failure.

Recommendation

The city should replace both with an accessible, perceptually ordered sequential scale, such as a light-to-dark or Viridis-style scale. Numerical labels and explicit risk thresholds should accompany the colour so that users can understand infection levels without depending on colour alone.

Q5. Visualizing 50,000 GPS Pings

A ride-hailing company has 50,000 GPS pings and wants to identify where demand concentrates. The three proposals have different strengths.

Option A – Plain scatter plot

A scatter plot preserves individual GPS locations and is therefore closest to the raw data. It can be useful when the analyst needs to inspect individual observations or unusual locations.

However, 50,000 points create severe overplotting. If 5,000 pings occur near one transport hub, many points will overlap. The viewer cannot easily tell whether a dark-looking area contains 500, 5,000 or 20,000 observations. The map may also become visually cluttered and slow to interact with.

Accuracy: high for individual locations, but weak for communicating density. Interpretability: low when the point count is extremely large.

Option B – 2D density contour map

A density contour map summarizes the concentration of points across geographic space. Areas with high demand appear as dense or closely spaced contours, allowing managers to identify hotspots quickly.

For example, if a railway station and an airport each have a large concentration of pings, the density map can show both as distinct hotspots even when thousands of individual points overlap.

The main limitation is smoothing. A bandwidth that is too large can merge two nearby demand centres; a bandwidth that is too small can create noisy, fragmented hotspots. The analyst should therefore test the bandwidth and compare it with known operational locations.

Accuracy: good for spatial concentration when the density method is appropriate. Interpretability: very high for hotspot identification.

Option C – Neighborhood bubble map

The bubble map aggregates the 50,000 observations by neighborhood and uses bubble area to show total demand. This is simple for management reporting. For example, a manager could immediately see that Neighborhood A has 8,000 pings, Neighborhood B has 5,000 and Neighborhood C has 2,000.

The disadvantage is loss of spatial detail. If Neighborhood A has one extremely busy street but the rest of the neighborhood is quiet, the single bubble hides that internal pattern. This is an aggregation problem and can be affected by the chosen neighborhood boundaries.

Accuracy: strong for neighborhood totals, weak for within-neighborhood patterns. Interpretability: high for summary decisions.

Decision matrix

- Plain scatter: preserves raw points, but severe overplotting at 50,000 observations.
- 2D density: best balance for discovering spatial hotspots, but depends on smoothing choices.
- Neighborhood bubbles: easiest for high-level reporting, but sacrifices fine spatial detail.

Final decision

I would approve the 2D density contour map for the primary operations dashboard. The operational question is not simply where individual GPS pings occurred; it is where demand concentrates. A density map directly answers that question while avoiding the major overplotting problem.

I would also add neighborhood boundaries and labels so managers can translate hotspots into operational areas. A secondary table or bubble view can provide exact neighborhood totals when resource allocation requires them.

Recommendation

Use the 2D density contour map as the primary dashboard view, supported by neighborhood-level totals as a secondary summary. Retain the raw scatter plot for analyst-level investigation rather than making it the main management visualization.

Overall Conclusion

Across all five cases, effective visualization depends on matching the representation to the decision and the audience. Probability should be framed in a way that people can understand; uncertainty should be visible without overwhelming mobile users; bar charts should respect proportional ink; colour should represent numerical order perceptually and accessibly; and large spatial datasets should be aggregated carefully to reveal patterns without hiding important information.

The common principle is to make the important information easier to perceive while avoiding visual features that introduce information that is not actually present in the data. A good visualization is therefore not simply attractive: it is accurate, interpretable, accessible and honest about uncertainty.